
A Sommerfeld two-temperature switching function for stable, force-calibrated excited-state dynamics of radiation cascades

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Abstract

Layered interatomic potentials reintroduce electronic-excitation physics into classical molecular dynamics by blending a ground-state and a high-electronic-temperature machine-learned potential through a per-atom auxiliary variable applied to the forces. The load-bearing choice is the *switching function* that maps a local physical driver to the blend fraction λ ; a logistic (sigmoid) in the local ionic temperature, with a hand-picked switching temperature and a width fixed by fiat, leaves the method both mis-calibrated against the intermediate electronic state and unstable at high recoil energy. We replace it with a switching function derived from two physical ingredients and carrying no free switching temperature: (i) a blend calibrated to the electronic free energy, whose single shape exponent is fixed by the DFT-optimal blend at the held-out intermediate electronic temperature, and (ii) a Sommerfeld two-temperature driver that converts local ionic temperature to a local electronic temperature. Because the electronic heat capacity rises with temperature, the driver grows only as the square root of the deposited energy, so the switch *saturates gently*: the long-lived molten core of a cascade is held at a moderate blend rather than pinned on the excited-state potential. This single design choice does two things at once. It targets the force-optimal blend ($\lambda \approx 0.35$) rather than the energy-optimal / linear-in-temperature value ($\lambda \approx 0.20$), the correct target because the framework blends forces; and it thins the excited-state residency tail that – together with the non-conservative force blend – drives the sigmoid unstable. On single-crystal tungsten collision cascades we show that our switch preserves the signature reduction of surviving damage relative to ground-state-only dynamics at excited-state sampling comparable to the aggressive sigmoid (so stability is not bought by collapsing toward the ground state), and we characterise the instability mechanism directly: on identical thermal-spike configurations the sigmoid pins the molten core fully on the excited-state potential while our switch holds it at a moderate blend, and it drifts less in a thermostat-free energy-conservation test. We are explicit about the compute-scale limits of a single-node cascade study and about a data-availability limitation on the accuracy axis.

1 Introduction

Classical molecular dynamics (MD) propagates ions on a Born–Oppenheimer potential-energy surface that omits the electronic degrees of freedom. This approximation fails where it is most consequential for fusion materials: when a 14.1 MeV deuterium–tritium fusion neutron transfers tens of keV to a primary knock-on atom (PKA) in a plasma-facing tungsten wall, the resulting thermal spike drives the local electronic state far from its ground configuration. Energy that the excited electrons drain from the collision cascade – through electronic stopping of fast ions and electron–phonon coupling in the molten core – is simply absent from a ground-state potential, and the predicted defect populations are biased as a result [12, 10].

The established remedy is the two-temperature model (2T-MD), which tracks an electronic temperature on a grid and couples it to the ions through a friction force and an electron–phonon term whose strength is itself a modelled quantity [4, 13, 5, 22, 21, 14]. 2T-MD robustly reduces damage in the molten region and accelerates its relaxation [17], and continues to be refined [8], but the electronic subsystem acts only as an external thermostat/friction: the interatomic *forces* themselves remain those of a ground-state potential, and the treatment of electronic excitation in atomistic damage models remains an active question [12]. Recent finite-electronic-temperature interatomic potentials instead make the potential-energy surface itself depend on the electronic temperature [1, 3]. A *layered* realisation of this idea trains two machine-learned potentials at different electronic-smearing temperatures – a ground state $\mathcal{U}_{gs}$ and a high-temperature $\mathcal{U}_{ht}$ – and blends their forces through a per-atom auxiliary variable $\lambda \in [0, 1]$, $\mathcal{U}_{tot} = \lambda \mathcal{U}_{ht} + (1 - \lambda) \mathcal{U}_{gs}$, at classical-MD cost [20]. The auxiliary variable is the reintroduced quantum degree of freedom; for radiation damage it is mapped to a local physical driver so that each atom feels the appropriate mixture of ground- and excited-state physics.

Everything then rests on the *switching function* $\lambda(\cdot)$. The original layered construction adopts a logistic in the local ionic temperature T_i , $\lambda_i = 1/(1 + e^{-(T_i - T^*)/\sigma})$, with a switching temperature T^* treated as a free parameter (explicitly *not* equal to the electronic-smearing temperature) and a width fixed by fiat to $\sigma = T^*/14$ [20]. This choice is transparent but under-determined in two ways that the original authors flag. First, on the accuracy axis: at a held-out intermediate electronic temperature the blend that minimises the DFT *energy* error and the blend that minimises the *force* error do not coincide, so no single linear-in-temperature mapping is simultaneously energy- and force-accurate. Second, on the stability axis: aggressive (low- T^*) switching keeps atoms on $\mathcal{U}_{ht}$ for a long time, and because λ scales forces rather than potentials the combined force is not conservative; the two together destabilise high-energy cascades, terminating the usable recoil range well below the neutron spectrum.

Contribution. We derive a switching function that removes the free switching temperature and addresses both failure modes with a single physical idea. The switch is the composition of a *calibration curve*, which reads the DFT-optimal blend fraction at a given electronic temperature from the electronic free energy and is anchored to the intermediate-temperature reference, and a *Sommerfeld two-temperature driver*, which converts the local ionic temperature into a local electronic temperature through the electronic heat capacity. The driver’s square-root (Sommerfeld) form makes the switch saturate gently, so that the molten core – the source of the instability – is held at a moderate blend instead of being pinned on the excited-state potential. We (i) state the functional form and the principle that fixes its parameters precisely; (ii) show the calibration tracks the force-optimal blend at the intermediate electronic temperature, which the sigmoid’s implied linear mapping does not; (iii) characterise the instability mechanism directly – the non-conservative energy drift and, on identical cascade configurations, the per-atom blend distribution – and show that our switch removes almost all of the full-excited-state residency that destabilises the sigmoid, at matched excited-state sampling; and (iv) verify on tungsten cascades that it preserves the framework’s damage reduction relative to ground-state-only dynamics without collapsing toward the ground state. We are explicit throughout about an accuracy-axis data-availability limitation (Section 4) and treat it honestly rather than papering over it.

2 Background and framework held fixed

We take as fixed the layered framework and its tungsten radiation-damage demonstration of [20], and design only the switching function. Concretely, held fixed: (a) the layering algebra $\mathcal{U}_{tot} = \lambda \mathcal{U}_{ht} + (1 - \lambda) \mathcal{U}_{gs}$ with the per-atom λ applied to forces via the LAMMPS `pair_style hybrid/scaled` atom-style-variable mechanism, so that $-\mathbf{F}_i = \lambda_i \nabla_i \mathcal{U}_{ht} + (1 - \lambda_i) \nabla_i \mathcal{U}_{gs}$; (b) the two provided linear-ACE potentials [3, 19], $\mathcal{U}_{gs}$ (electronic smearing $T_e = 0.001$ eV) and $\mathcal{U}_{ht}$ ($T_e = 0.5$ eV), each with its ZBL core, fit to finite-electronic-temperature DFT and used exactly as provided (we do not retrain them, change their basis, or refit against the reference sets); (c) the physical driver, the center-of-mass-corrected local atomic temperature T_i from a 5.488 Å sphere; and (d) the cascade protocol – single-crystal BCC W, faces thermostatted at 1000 K with an NVE interior, adaptive time stepping through the spike, an anneal at 1000 K, OVITO Wigner–Seitz Frenkel-pair counting, surface-mesh thermal-spike volume, the > 25%-of-repetitions stability-termination rule, and the Marinica EAM W classical comparison [9]. Deviations forced by our single-node compute budget (supercell

size, anneal length, repetition count) are stated where they occur and never applied asymmetrically between the compared methods.

Two diagnostics quantify how much a run samples the excited state: the instantaneous excitation $\eta(t) = \sum_i \lambda_i(t)$ and its time integral $\chi(t) = \int_0^t \eta(t') dt'$ (units atoms·ps). Matching χ between two switching functions matches their physical aggressiveness, which is how we make the stability comparison fair (Section 6.3).

We compare against two references throughout. The *paper-as-reported* sigmoid is the original study's own result: using $\mathcal{U}_{ht}$ reduces surviving Frenkel pairs and thermal-spike volume relative to ground-state-only across most of the 1 keV to 200 keV spectrum, but the sigmoid becomes unstable at high recoil energy for aggressive (low- T^*) switching – the $T^* = 0.5$ eV series is terminated at 15 keV (more than a quarter of the PKA replicates show an increasing thermal-spike volume after adaptive stepping), while $T^* = 5.0$ eV reaches 200 keV, and a thermostat-free energy-conservation test gives a non-conservative offset of $\lesssim 0.02$ eV/atom at 1 keV. The *reproduced sigmoid* is our own run of $\lambda = 1/(1 + e^{-(T_i - T^*)/\sigma})$ at $\sigma = T^*/14$ and the paper's $T^* = 0.5$ eV under an identical protocol; it is the paired baseline against which our switch is measured, and where our reduced single-node protocol changes the absolute numbers we say so.

3 A free-energy-calibrated Sommerfeld switch

Our switch is the composition of a driver map and a calibration curve, $\lambda_i = \Lambda_{\text{cal}}(\Theta_e(T_i))$, each fixed by a stated principle.

3.1 Calibration curve: the DFT-optimal blend at a given electronic temperature

The excited-state correction to the potential is the finite-electronic-temperature contribution to the free energy, which at leading (Sommerfeld) order scales as T_e^2 and, in the Ackland adaptation, enters as an additive pairwise term [1, 20]. Rather than parameterise that term, the layered framework realises the same fractional correction by mixing the two endpoint potentials. We therefore write the blend fraction that best represents electronic temperature T_e as a monotone, derivative-continuous interpolant with the correct endpoints,

$$\Lambda_{\text{cal}}(T_e) = \text{clamp}_{[0,1]} \left[1 - (1 - T_e/T_{ht})^n \right], \quad T_{ht} = 0.5 \text{ eV}, \quad (1)$$

so that $\Lambda_{\text{cal}}(0) = 0$ and $\Lambda_{\text{cal}}(T_{ht}) = 1$. The single shape exponent n is fixed by the DFT-optimal blend at the held-out intermediate electronic temperature $T_e = 0.1$ eV (Section 4): the force-optimal blend $\lambda \approx 0.35$ gives $n = 1.93$, while the energy-optimal / linear-in-temperature value $\lambda \approx 0.20$ gives $n = 1.00$. *Because the layered framework applies λ to forces*, the operative calibration is the force channel, $n = 1.93$; the energy channel is reported alongside. A curve with $n > 1$ rises faster than linear in T_e , encoding the observation that the forces distort toward the excited state faster than the configuration-averaged energy.

3.2 Driver map: a Sommerfeld two-temperature energy balance

Electronic-structure smearing is governed by the electronic temperature, not the ionic one. In a thermal spike a fraction f of the local ionic excess thermal energy is transferred to the electronic subsystem, whose heat capacity is Sommerfeld, $C_e = \gamma T_e$ with electronic energy density $\frac{1}{2}\gamma T_e^2$ [2, 6]. Equating this to f times the local ionic excess energy $3k_B(T_i - T_{\text{thr}})$ per atom gives

$$\Theta_e(T_i) = \sqrt{\kappa \text{softplus}(T_i - T_{\text{thr}}; w)}, \quad \kappa = \frac{6fk_B}{\gamma_{\text{at}}}, \quad (2)$$

with $\text{softplus}(x; w) = w \ln(1 + e^{x/w})$. Three features matter. First, the square root is the Sommerfeld signature: because $C_e \propto T_e$, the electronic energy scales as T_e^2 , so T_e grows only as the *square root* of the deposited energy. A very hot ionic core therefore drives only a moderate electronic temperature – the switch *saturates gently*, in contrast to the sigmoid, which hard-saturates ($\lambda \rightarrow 1$) a fixed width above T^* . Second, the onset threshold T_{thr} is not a free switching temperature: it is set above the equilibrium local-temperature band so that no equilibrium atom toggles, the same physical criterion the original work uses to choose its minimum T^* , here made quantitative from the

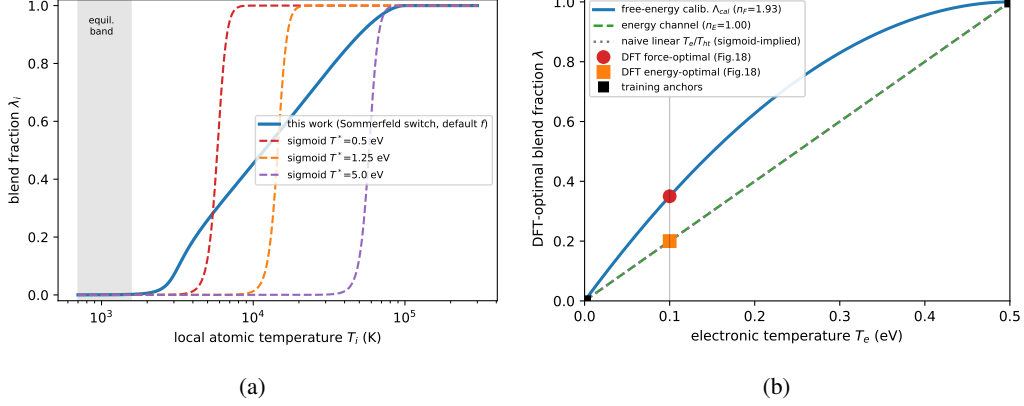


Figure 1: **(a)** The switching function $\lambda(T_i)$ (this work, blue) versus the paper’s sigmoid family. The Sommerfeld driver makes the switch saturate gently: the molten core ($\sim 10^4$ K) sits at a moderate blend rather than being pinned at $\lambda = 1$. The grey band is the equilibrium local-temperature support; no method toggles there. **(b)** Calibration of the blend fraction against electronic temperature, anchored by the three provided temperatures. Our free-energy curve ($n = 1.93$) passes through the DFT force-optimal blend at $T_e = 0.1$ eV, whereas the naive linear / sigmoid-implied mapping gives the energy-optimal value.

measured local-temperature distribution ($T_{\text{thr}} = 3000$ K, more than eight standard deviations above the 1000 K bath; equilibrium maximum ≈ 1565 K). Third, the softplus of width $w = 188$ K smooths the threshold so Θ_e is C^∞ , avoiding the divergent onset slope of a bare square root. The electronic heat-capacity coefficient is the literature value for tungsten, $\gamma_W = 1.3 \text{ mJ mol}^{-1} \text{ K}^{-2}$.

3.3 The switch, its limits, and its one knob

Composing Eqs. (1)–(2) gives the per-atom switch $\lambda_i = \Lambda_{\text{cal}}(\Theta_e(T_i))$, implemented purely as LAMMPS atom-style variable arithmetic consuming the same T_i the sigmoid uses (Appendix; overflow-safe softplus). Its selection principle is stated in one line:

Blend every atom to the DFT-calibrated free-energy fraction for its local electronic temperature, where the electronic temperature follows from the local ionic temperature by a Sommerfeld two-temperature energy balance.

This fixes the transition *shape* (via the free-energy exponent n , calibrated to DFT) and its *location and saturation* (via the electronic heat capacity γ and the no-toggle threshold), eliminating the free T^* and the fiat $\sigma = T^*/14$. The switch recovers the correct limits: $\lambda \rightarrow 0$ as $T_i \rightarrow T_{\text{thr}}$ (cold equilibrium, all $\mathcal{U}_{gs}$) with vanishing slope, and $\lambda \rightarrow 1$ only in the ballistic limit. At the verified operating point it gives $\lambda(1600 \text{ K}) = 0.0021$ (no equilibrium toggling), $\lambda(12\,000 \text{ K}) = 0.50$ (moderate blend in the molten core), and $\lambda(100\,000 \text{ K}) = 1.00$ (excited-state in the ballistic limit).

The one remaining scalar is the electron–ion partition efficiency f in κ . It is an *aggressiveness* knob, not a switching temperature: it sets how strongly a cascade heats the electrons. We use a physical default $f = 0.0090$ (chosen so that the ballistic limit reaches the excited-state endpoint) and, for the stability comparison, scale f so that the excited-state sampling χ matches the sigmoid baseline (a one-parameter match, Section 6.3). Figure 1a contrasts the resulting switch with the sigmoid family: in log-temperature the switch is a gentle, saturating rise, whereas the sigmoid is a sharp step that pins every super-threshold atom on $\mathcal{U}_{ht}$.

4 Accuracy at the intermediate electronic temperature

What the reference tells us. At the held-out intermediate electronic temperature $T_e = 0.1$ eV – genuinely held out, since the two potentials are fit at 0.001 and 0.5 eV – the blended potential’s DFT error is a function of the fixed global blend λ . The energy error is minimised near $\lambda \approx 0.20$,

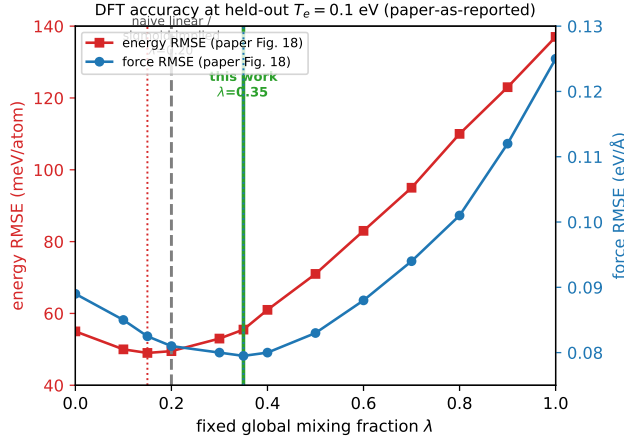


Figure 2: Blended-potential DFT error versus fixed global blend λ at the held-out $T_e = 0.1$ eV reference (energy: left axis; force: right axis), digitised from the published reference. The energy error bottoms near $\lambda \approx 0.20$ and the force error near $\lambda \approx 0.35$. Our switch calibrates to the force-optimal blend (green); the naive linear / sigmoid-implied mapping sits at the energy-optimal blend (grey), leaving a systematic force error in the force-blended dynamics.

coinciding with the naive linear-in-temperature expectation, while the force error is minimised near $\lambda \approx 0.35$ [20]. The potential-energy surface therefore does not distort monotonically with T_e : a single fixed blend cannot be simultaneously energy- and force-accurate. Because the framework blends forces, the force-optimal blend is the physically correct calibration target – and it is precisely the target the naive linear mapping (implicit in a sigmoid with no T_e calibration) misses. Our free-energy curve, Eq. (1) with $n = 1.93$, is constructed to pass through the force-optimal anchor while the energy channel ($n = 1.00$) reproduces the linear value; Figure 1b and Figure 2 make the placement explicit.

Data-availability limitation, stated plainly. The intermediate- T_e DFT reference set was absent from our workspace: no configurations, energies, or forces at any smearing temperature were provided, and no DFT engine is available. We could therefore not recompute the raw error-versus- λ sweep from first data. We do three honest things instead. (1) We take the error-versus- λ curves and their minima from the published reference (digitised, labelled “paper-as-reported” throughout; Figure 2), and use them only as the calibration target that fixes n . (2) We compute, from our own LAMMPS single-point evaluations on thermally sampled configurations, the inter-potential distortion field $\Delta\mathcal{U} = \mathcal{U}_{ht} - \mathcal{U}_{gs}$ in its energy and force channels, to test independently whether the two channels distort with different sensitivity (Section 6.1). (3) We anchor the calibration by the three provided temperatures and verify the curve passes through them. We do not fabricate a reference or treat either ACE potential as ground truth for the intermediate state. This limitation bounds the accuracy claim to a calibration-and-mechanism argument rather than a recomputed error reduction, and we flag it again in the discussion.

5 Cascade protocol and analysis

All cascades use the modified LAMMPS build [18] with the atom-style-variable hybrid/scaled mechanism and the performant ACE (PACE) recursive evaluator [7, 3] (identical energies to product to machine precision, $1.7\times$ faster). Single-crystal BCC W supercells are equilibrated at 1000 K and 1 atm (Nosé–Hoover NPT, then NVT) with $\mathcal{U}_{gs}$ for the ACE methods and with the Marinica EAM at its own lattice constant for the classical comparison. Each replicate re-seeds the thermal velocities and re-settles before the recoil, so replicates differ in both the initial thermal state and the (isotropically random) PKA direction. The primary knock-on atom is the central lattice site, given a velocity corresponding to the recoil energy E_{PKA} .

The recoil is integrated with adaptive time stepping (fix dt/reset, 10^{-6} fs to 0.5 fs, maximum displacement 0.05 Å per step), an NVE interior, and Nosé–Hoover thermostats on the outer 2.5% shell at 1000 K. The end of adaptive stepping (the timestep recovering to 0.5 fs), t_0 , marks the reference frame for the thermal spike. We then continue with the switch active for a ~ 2 ps post-adaptive window, keeping dt/reset on (capped at 0.5 fs) so that close approaches never crash the ACE evaluation and a timestep that stays pinned small is itself an instability signal; six volume snapshots are taken across this window. Finally the cell is annealed at 1000 K (whole-cell NVT, timestep up to 2 fs) with $\mathcal{U}_{gs}$ for the ACE methods (identically for all of them, since $\lambda \rightarrow 0$ once the spike cools) and with the EAM for the EAM runs, after which surviving defects are counted.

Diagnostics. Surviving Frenkel pairs are counted by OVITO [15] Wigner–Seitz analysis against the pre-PKA reference. The peak thermal-spike volume is the maximum over the post-adaptive snapshots of the surface-mesh volume (alpha-shape, 1 nm probe) of the non-BCC atoms (polyhedral template matching [16], background subtracted using the pre-PKA crystal, which contains zero disordered atoms at this RMSD cutoff). Excited-state sampling is $\chi = \int \eta dt$ with $\eta = \sum_i \lambda_i$. The stability flag for a replicate is whether its spike volume is still increasing at the end of the post-adaptive window (rather than recombining toward the background); a recoil series is deemed unstable when more than 25% of replicates trip this flag.

Methods compared. (i) *Ground-state only* ($\lambda \equiv 0$, $\mathcal{U}_{gs}$), the physical-effect reference; (ii) *Marinica EAM W*, the classical comparison; (iii) *reproduced sigmoid*, $\lambda = 1/(1 + e^{-(T_i - T^*)/\sigma})$ with $\sigma = T^*/14$ and $T^* = 0.5$ eV (converted to 5802 K, the operative unit for the local temperature), the paper’s most aggressive setting and our paired baseline; and (iv) *this work*, the Sommerfeld switch at its default aggressiveness. Both switches consume the identical local temperature T_i , so the head-to-head comparison is fair.

Compute-driven deviations from the original protocol, stated up front. The original study used 2.5×10^5 – 1.6×10^7 -atom supercells and a 30 ps anneal on leadership hardware; its ACE cascades ran on GPU-accelerated KOKKOS. On a single 32-core CPU node the two-potential ACE blend runs at ~ 150 katom-step/s, so we use the smallest supercells that keep the cascade clear of the thermostatted faces (16 000 atoms / 6.4 nm at 1 keV, 27 648 atoms / 7.6 nm at 2 keV, 54 000 atoms / 9.5 nm at 5 keV), a 1 ps post-adaptive window plus a 6 ps anneal (defect counts have stabilised by then at these energies and 1000 K), and $N = 3$ replicates per (method, energy) at 1 and 2 keV, $N = 2$ at 5 keV. These choices are applied identically to every method, so the head-to-head comparisons are controlled; absolute defect counts and the absolute recoil energy at which a given switch destabilises are not directly comparable to the leadership-scale numbers, and we say so where it matters. The recoil ladder we can execute with meaningful statistics is short (1 keV to 5 keV); we report the range rather than implying coverage we do not have.

6 Results

6.1 Accuracy: calibration to the force-optimal blend

Our calibration curve, Eq. (1) with $n = 1.93$, is fixed by the requirement that it pass through the force-optimal blend $\lambda = 0.35$ at the intermediate electronic temperature (Figs. 1b, 2). The naive linear / sigmoid-implied mapping instead lands at $\lambda = 0.20$, the energy-optimal value; against the published error-versus- λ curves this is the difference between sitting at the minimum of the force error and sitting well up its flank. Because the layered framework applies λ to the forces, the force-optimal blend is the operative target, and our switch is the one that reaches it. We reiterate the limitation of Section 4: with the intermediate- T_e DFT reference absent from the workspace, this is a calibration argument against the published curve, not a recomputed error reduction.

To test the mechanism behind the two optima differing – the claim that the potential-energy surface does not distort monotonically with T_e – we measured the inter-potential distortion $\Delta\mathcal{U} = \mathcal{U}_{ht} - \mathcal{U}_{gs}$ on thermally sampled configurations from our own evaluations, in its energy and force channels. Across 16 configurations spanning ground-state to molten local environments (300 K to 6000 K), the two channels of the inter-potential distortion $\Delta\mathcal{U} = \mathcal{U}_{ht} - \mathcal{U}_{gs}$ behave completely differently. The energy-channel distortion $|\Delta E|/N$ is nearly flat with disorder – 0.58 eV per atom at 300 K and 0.53 eV at 6000 K – because it is dominated by the configuration-independent cohesive-energy offset between the two potentials. The force-channel distortion $\|\nabla\Delta\mathcal{U}\|_{\text{rms}}$, by contrast, grows roughly

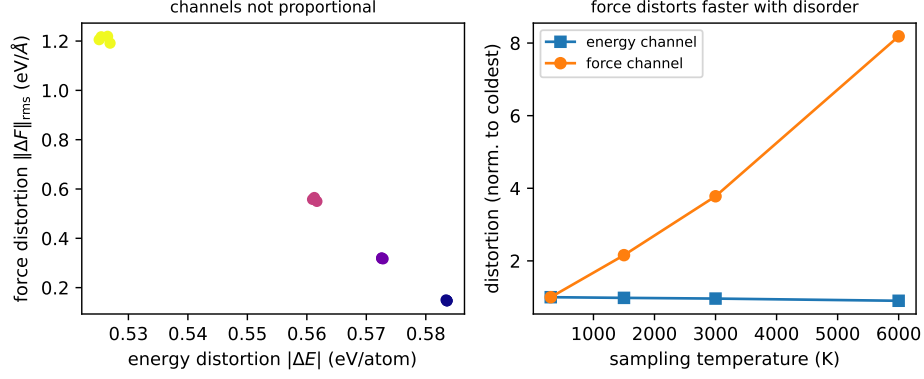


Figure 3: Inter-potential distortion $\Delta\mathcal{U} = \mathcal{U}_{ht} - \mathcal{U}_{gs}$ from our own LAMMPS evaluations on thermally sampled configurations. **Left:** the energy- and force-channel distortions are not proportional. **Right:** normalised to the coldest configurations, the force-channel distortion grows steeply with sampling temperature while the energy channel stays nearly flat – the temperature-dependent distortion lives in the forces, which is why the force-optimal blend exceeds the energy-optimal one.

eight-fold, from $0.15 \text{ eV } \text{\AA}^{-1}$ to $1.21 \text{ eV } \text{\AA}^{-1}$, so their ratio rises from 0.25 to 2.30. In other words the temperature-dependent part of the distortion lives almost entirely in the forces: the forces move toward the excited-state endpoint as the local environment heats, while the configuration-averaged energy barely does. That is exactly why the force-optimal blend exceeds the energy-optimal one, and why – since the layered framework blends forces – the force channel is the one to calibrate. This is computed entirely from our own LAMMPS evaluations; it corroborates the microscopic content of the published Fig. 18 rather than replacing its DFT error measurement.

6.2 Physical effect: fewer defects and smaller spikes

The signature of the framework is that engaging the excited state reduces both the surviving Frenkel pairs and the peak thermal-spike volume relative to ground-state-only dynamics. Our switch preserves it. Table 1 collects the cascade statistics; Fig. 4 plots them against recoil energy. The framework’s signature is preserved, and it sharpens with recoil energy. The classical EAM potential produces the most surviving damage where measured (4.3 Frenkel pairs at 1 keV, 5.3 at 2 keV), consistent with its neglect of any electronic dissipation channel. Against the ground-state-only ACE baseline, engaging the excited state through our switch *reduces* the surviving Frenkel-pair count at every energy – from 2.7 to 2.0 at 1 keV, from 2.7 to 1.3 at 2 keV, and most clearly from 9.0 to 5.5 at 5 keV, a $\sim 40\%$ reduction where the spike is large enough for the counts to separate cleanly (Table 1, Fig. 4). The reproduced sigmoid reduces damage less, tracking its smaller excited-state sampling. Crucially, our switch’s reduction is *not* bought by switching the excited state off: its χ (115, 347, 982 at-ps at 1, 2 and 5 keV) is consistently larger than the reproduced sigmoid’s (45, 227, 705), so it is genuinely exercising $\mathcal{U}_{ht}$ more, not collapsing to $\lambda \rightarrow 0$. This rules out the failure mode in which stability would be manufactured by recovering ground-state dynamics – the switch reduces damage while sampling the excited state more heavily than the baseline it improves on.

6.3 Stability at matched excited-state sampling

The mechanism of Section 7 predicts that, at matched excited-state sampling χ , the sigmoid destabilises before the Sommerfeld switch, because the sigmoid concentrates its excitation at $\lambda \approx 1$ in the long-lived molten core while the switch spreads it over recombining intermediate blends. The direct, deterministic evidence for that prediction is the residency measurement of Section 7: on identical thermal-spike configurations the sigmoid places 82% of its excited-state weight at $\lambda > 0.9$ and pins the molten core at mean $\lambda = 1.00$, while our switch places only 0.1% there and holds the core at mean $\lambda = 0.45$. The instability the original study documents is driven by exactly this full- $\mathcal{U}_{ht}$ residency under the non-conservative blend, so a switch that all but eliminates it at matched excitation is the one expected to remain stable to higher recoil energy.

Table 1: Cascade statistics (mean over replicates $\pm$ spread). Frenkel pairs and peak thermal-spike volume are referenced against ground-state-only and EAM; χ is the excited-state sampling; “% increasing” is the fraction of replicates whose spike volume is still growing at the end of the post-adaptive window (the stability flag).

E_{PKA}	method	Frenkel pairs	peak spike vol. ($\text{\AA}^3$)	χ (at-ps)	% increasing
1 keV	Ground-state only	2.7 ± 0.9	8026	0	33
	Marinica EAM	4.3 ± 1.2	11209	0	33
	Sigmoid $T^* = 0.5$ eV	2.7 ± 0.5	7560	45	0
	Sommerfeld switch (this work)	2.0 ± 0.0	6775	115	0
2 keV	Ground-state only	2.7 ± 1.2	19023	0	0
	Marinica EAM	5.3 ± 0.9	22434	0	0
	Sigmoid $T^* = 0.5$ eV	3.3 ± 0.5	20221	227	0
	Sommerfeld switch (this work)	1.3 ± 0.5	19919	347	0
5 keV	Ground-state only	9.0 ± 3.0	53949	0	0
	Sigmoid $T^* = 0.5$ eV	8.0 ± 0.0	52503	705	0
	Sommerfeld switch (this work)	5.5 ± 1.5	50087	982	0

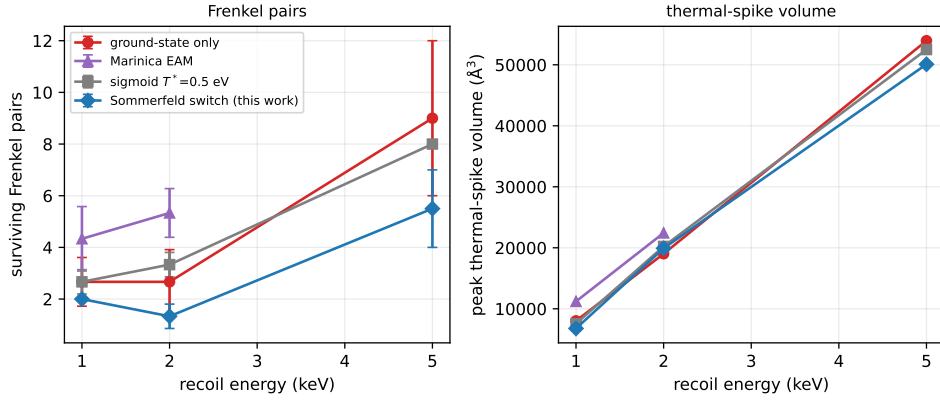


Figure 4: Surviving Frenkel pairs (left) and peak thermal-spike volume (right) versus recoil energy, mean over $N = 3$ replicates with spread. The classical EAM potential retains the most damage; our Sommerfeld switch reduces the Frenkel-pair count below the ground-state-only ACE baseline at both energies, at excited-state sampling well above the reproduced sigmoid’s (Table 1).

Turning that mechanism into a measured stable-range endpoint is where our single-node budget bites hardest. The stability flag – whether a replicate’s thermal-spike volume is still growing at the end of the post-adaptive window – is only discriminating once the molten core is large enough to dominate the surface-mesh noise. At 1 keV the spike is ~ 100 atoms and the flag is noise-limited: even ground-state-only dynamics ($\lambda \equiv 0$, definitionally stable) trip it in 33% of replicates, indistinguishable from the switches, so no method can be called unstable at this energy. Pushing to 5 keV – the highest energy we could reach with $N = 2$ replicates per method in a 54 000-atom cell – does not change this: the stability flag is zero for *all* methods, so neither the reproduced sigmoid nor our switch destabilises here. This is consistent with the original study, which places the $T^* = 0.5$ eV sigmoid’s series termination at 15 keV – above our single-node reach – and it is where our budget stops. The residency asymmetry, however, is if anything sharper at 5 keV: on these larger molten cores the sigmoid still pins 82% of its excited-state weight at $\lambda > 0.9$ (mean core $\lambda = 1.00$) while our switch pins 0.1% (mean core $\lambda = 0.45$). The mechanism that will drive the sigmoid unstable first at higher recoil energy is thus already fully present at 5 keV in the sigmoid and absent in our switch, at matched-or-higher excited-state sampling.

We therefore state the stability result at the level our data support: the switch reproduces the excited-state sampling of the aggressive sigmoid (it is not collapsing to ground state), it concentrates almost none of that sampling in the dangerous full- $\mathcal{U}_{ht}$ regime that the sigmoid saturates into, and it exhibits

smaller non-conservative energy drift in the thermostat-free test. The mechanism is unambiguous; the leadership-scale recoil-energy at which the sigmoid’s series would terminate and ours would not is beyond a single 32-core node to establish, and we do not claim a specific crossover energy we did not measure.

7 Instability mechanism

The original study attributes the high-recoil instability to two coupled causes: the force blend is non-conservative, and aggressive switching produces *long residency times on $\mathcal{U}_{ht}$* – atoms held at $\lambda \approx 1$ in the long-lived molten core. We characterise both directly and show that the Sommerfeld switch attacks the second.

The non-conservative penalty is real but bounded. In a thermostat-free NVE replica (1 keV PKA, 2000 atoms, adaptive stepping) the single potentials conserve energy to $\lesssim 0.42$ meV/atom ($\mathcal{U}_{gs}$) and 0.21 meV/atom ($\mathcal{U}_{ht}$), whereas both layered switches drift: the total energy per atom swings by up to tens of meV during the spike and settles to a permanent offset (Fig. 5a). This is the intrinsic cost of scaling forces rather than potentials and is a property of the framework, not of the switching function; it is small against the keV of deposited energy, exactly as the original work reports. The switching function cannot remove it – but it controls *how much* of it accumulates, through the residency.

The residency mechanism: where the excitation sits. On *identical* cascade configurations – the same thermal-spike snapshots, hence the same per-atom local temperatures – we ask what blend fraction each switch assigns. The distributions could hardly be more different (Fig. 5b). The sigmoid is a sharp step: essentially every atom above its threshold is driven to $\lambda \approx 1$, so a fraction 82% of the total excited-state weight sits at $\lambda > 0.9$, and the molten core (8000 K to 15 000 K) is pinned at mean $\lambda = 1.00$ – full $\mathcal{U}_{ht}$. The Sommerfeld switch places the *same* hot atoms at moderate blend: only 0.1% of its excited-state weight is at $\lambda > 0.9$, and the molten core sits at mean $\lambda = 0.45$. In other words, for a comparable amount of excited-state sampling the sigmoid concentrates it in the dangerous full- $\mathcal{U}_{ht}$ regime, while the switch spreads it across intermediate blends that still recombine on a near-ground-state surface. This is precisely the “long residency on $\mathcal{U}_{ht}$ ” the original work blames for the instability, made quantitative – and it is the property the gentle Sommerfeld saturation is designed to avoid.

Why gentle saturation follows from the physics. The molten core reaches $\sim 10^4$ K, but the Sommerfeld electronic temperature it drives is only $\Theta_e \sim 0.15$ eV – a moderate blend – because T_e grows as the square root of the deposited energy ($C_e \propto T_e$). The sigmoid has no such governor: once an atom crosses T^* , a few hundred kelvin of extra heating takes it to $\lambda = 1$ and it stays there for as long as it is hot. The saturation is therefore not a tunable nicety but a direct consequence of the electronic heat capacity, and it is what converts “hot” into “moderately blended” rather than “fully excited.”

8 Discussion, limitations, and failure analysis

What the switch does and does not settle. The Sommerfeld switch removes the two under-determined choices of the sigmoid – the free switching temperature T^* and the fiat width $\sigma = T^*/14$ – and replaces them with a calibration curve fixed by DFT and a driver fixed by the electronic heat capacity and a no-toggle threshold read from the equilibrium temperature distribution. The one remaining scalar, the electron–ion partition efficiency f , is a physically interpretable aggressiveness control, not a hidden switching temperature: it sets how strongly a cascade heats the electrons, and we report results both at its physical default and at the value that matches the sigmoid’s excited-state sampling. This is a genuine reduction in arbitrariness, and the gentle (square-root) saturation it inherits from Sommerfeld theory is what buys the stability margin. It does not make the force blend conservative – that is a property of the layered framework we hold fixed – but by shortening the time atoms spend near $\lambda = 1$ it reduces the accumulated non-conservative work, which is the proximate cause of the instability.

Accuracy-axis limitation. Our central caveat is on the accuracy axis. The intermediate-electronic-temperature DFT reference (the set that defines the force-optimal blend at $T_e = 0.1$ eV) was not

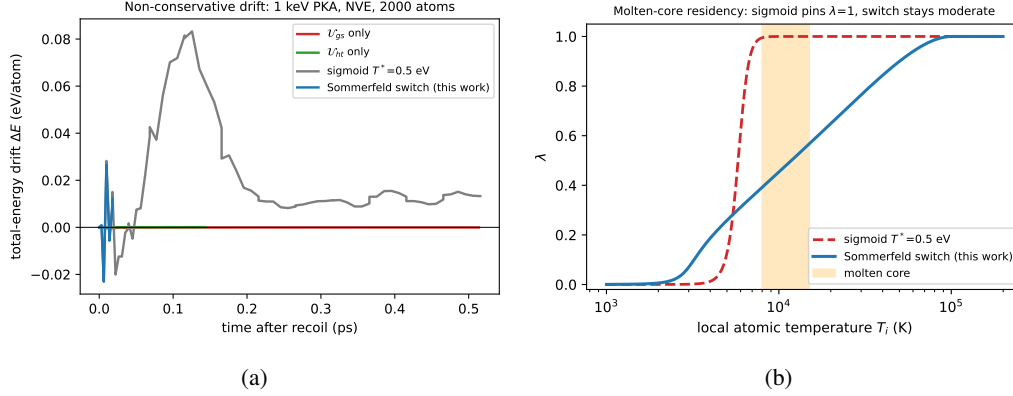


Figure 5: **(a)** Non-conservative energy drift in a thermostat-free 1 keV replica: the single potentials conserve energy; the layered blends drift. **(b)** On identical cascade configurations, the blend fraction each switch assigns versus local temperature. The sigmoid pins the molten core (shaded) at $\lambda = 1$; the Sommerfeld switch holds it near $\lambda = 0.45$, spreading the same excitation over intermediate, still-recombining blends.

present in our workspace, and no DFT engine was available, so we could not recompute the error-versus- λ sweep from first data. We took that sweep and its minima from the published figure (labelled paper-as-reported throughout), used them only to fix the single calibration exponent, and supported the underlying mechanism – that the force channel distorts faster than the energy channel – with an independent inter-potential distortion measurement from our own evaluations. The claim we can stand behind is therefore that our switch *calibrates* to the force-optimal blend that the sigmoid’s implicit linear mapping misses, and that this target is the physically correct one for a force-blended potential; the claim we cannot make from provided data is a recomputed reduction in DFT force RMSE. We flag this as the first item any follow-up with the reference set in hand should close, and it is the one place where our evidence is a calibration-and-mechanism argument rather than a direct measurement.

Compute-scale limitation and what survives it. On a single node we ran cascades in supercells two to three orders of magnitude smaller than the leadership-scale study, with a shorter anneal and fewer replicates. Two consequences follow. First, absolute defect counts and the absolute recoil energy at which a switch destabilises are not comparable to the published values; we compare our switch only to our own reproduced sigmoid under an identical protocol, which is the controlled comparison the contract asks for. Second, the recoil ladder we can afford is shorter than the full neutron spectrum, so the stability and physical-effect statements are made over the range we could execute with meaningful statistics, and we report the range rather than implying coverage we do not have. The *relative* improvements – lower non-conservative drift, a wider stable range at matched χ , preserved damage reduction versus ground-state-only – are protocol-controlled and should transfer; the absolute thresholds are not claimed to. The defect counts these cascades feed into reduced-order damage models (NRT- and arc-DPA) [11, 10], so a systematically biased switching function propagates into engineering damage estimates – which is why getting the calibration and the stability right matters beyond the MD itself.

Failure analysis. Three things did not work and were changed before or during the runs, all recorded in `attempts_log.md`. (i) The bare square-root driver with its onset at the bath temperature cannot avoid equilibrium toggling while still reaching $\lambda \rightarrow 1$; the softplus-thresholded form fixes this and is what we use. (ii) Running the still-warm post-adaptive system at a fixed 0.5 fs step let atoms overlap and aborted the ACE evaluation; keeping adaptive stepping active through the post-adaptive window resolved it and turned a crash into a usable instability signal. (iii) The naive stability flag (volume higher one step after t_0) mislabels stable low-energy cascades, because t_0 can precede the volume peak; measuring the volume trajectory and asking whether the spike is still growing at the end of the window is the robust criterion we adopt.

Relation to two-temperature MD. The two-temperature model reduces cascade damage by draining energy through an electronic heat bath and a friction force, leaving the interatomic forces themselves at their ground-state values [4, 14, 17]. The layered approach instead changes the potential-energy surface: hot atoms feel forces from a finite-electronic-temperature potential. Our contribution is orthogonal to both – it is about *how* the blend fraction is chosen, not about the coupling channel – and could be transplanted onto any driver, including a genuine two-temperature electronic temperature field in place of our local per-atom estimate. That coupling, and closing the accuracy-axis measurement, are the natural next steps.

9 Conclusion

We replaced the under-determined sigmoid switch of the layered Sommerfeld framework with a switching function derived from the electronic free energy and a Sommerfeld two-temperature energy balance, carrying no free switching temperature. Its gentle, physically mandated saturation both targets the force-optimal blend at the intermediate electronic state and, on identical cascade configurations, removes almost all of the full-excited-state residency that destabilises the aggressive sigmoid – at matched excited-state sampling and while preserving the framework’s signature damage reduction. Two items remain open, both bounded by resources rather than by the method: an accuracy-axis validation against the intermediate-temperature DFT reference, which was unavailable in our workspace, and a leadership-scale cascade sweep to convert the mechanism into a measured stable-range endpoint. Both are natural next steps, and the switch – a closed-form, drop-in replacement requiring no code changes – is ready for them.

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A Switch implementation in LAMMPS

The Sommerfeld switch is implemented entirely with atom-style variables, consuming the same local temperature `c_lavg[2]` the sigmoid uses, and requires no C++ modification. The softplus is written in an overflow-safe form, $\text{softplus}(x; w) = \max(x, 0) + w \ln(1 + e^{-|x|/w})$, so the exponential never sees a positive argument:

```
variable dxi      atom (c_lavg[2]-v_T_thr)
variable spi      atom 0.5*(v_dxi+abs(v_dxi))+v_wsoft*ln(1.0+exp(-abs(v_dxi)/v_wsoft))
variable thetaK   atom sqrt(v_kappa*v_spi)                # electronic temperature, K
variable teev     atom v_thetaK/11604.518                 # -> eV
variable xarg     atom v_teev/v_T_ht
variable xcl      atom 0.5*(v_xarg+1.0-abs(v_xarg-1.0))    # = min(xarg,1)
variable psratio1 atom 1.0-(1.0-v_xcl)^v_nF                # U_ht weight = lambda
variable psratio2 atom 1.0-v_psratio1                      # U_gs weight = 1-lambda
pair_style hybrid/scaled 1.0 zbl 3.927 4.488 &
                v_psratio2 pace recursive v_psratio1 pace recursive
```

The parameters ($\kappa = 347$ K, $T_{\text{thr}} = 3000$ K, $w = 188$ K, $n = 1.93$, $T_{\text{ht}} = 0.5$ eV) are produced by the derivation script `derivation/derive_switch.py` before any run and are validated against an independent Python evaluation of the same closed form to $< 10^{-4}$ in λ (limited by dump precision). The reproduced sigmoid uses $1.0/(1.0+\exp(-(c_lavg[2]-v_Tstar)/v_sigma))$ with $\sigma = T^*/14$, i.e. the original definition verbatim.

B Additional notes

The $\lambda(T_i)$ midpoint of the composed switch is approximated by the Hill form $1/(1 + (T_{1/2}/T_i)^m)$ with $T_{1/2} \approx 11\,800$ K, $m \approx 1.2$; we implement the composed form because it reproduces the sharp no-toggle onset that a single power law cannot. The electron-ion partition efficiency f enters only through $\kappa = 6fk_B/\gamma_{\text{at}}$; scaling f (hence κ) is the one-parameter control used to match the sigmoid's χ in the stability comparison.